A1 production kernel manifest – non-vacuous consistency gates (v1.1)

TECT verification-first repository · claim A1-PRODUCTION-KERNEL-MANIFEST

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1. Purpose and scope

A production config may be cited as the numerical implementation of A2/A3 only if its stored kernel parameters are mutually consistent AND it lies in the analytic branch. The v1.0 gate set contained a vacuity defect: with $m_{\mathrm{sh}}^2 := r - Z^2/(4Y)$ and $q_\star^2 := -Z/(2Y)$,

$$r - (m_{\mathrm{sh}}^2 + Y q_\star^4) \equiv 0 \qquad \text{for all } (r, Z, Y), \tag{1}$$

so that gate auto-passed and tested nothing. v1.1 corrects this by gating the *independent stored fields* ($\texttt{r\_zero}, \texttt{mu2\_shell}, \texttt{q0}$) against the identity, which is non-vacuous. Verification: `codes/foundations/a1_kernel_checks.py` v1.1 (7/7 PASS).

2. The v1.1 schema (non-vacuous)

A config stores three independent fields $\texttt{r\_zero} := D(0)$, $\texttt{mu2\_shell} := D(q_\star)$, $\texttt{q0} := q_\star$ together with (Z, Y). Define

$$\delta_Z := \left|Z + 2Y\,\texttt{q0}^2\right|, \quad \delta_m := \left|\texttt{mu2\_shell} - \left(\texttt{r\_zero} - \tfrac{Z^2}{4Y}\right)\right|, \quad \delta_r := \left|\texttt{r\_zero} - \left(\texttt{mu2\_shell} + Y\,\texttt{q0}^4\right)\right|, \tag{2}$$

with *separate* tolerances $\varepsilon_Z, \varepsilon_m, \varepsilon_r$ (the residuals carry different dimensions and must not be merged). The legacy alias $\texttt{mu2} = r$ is FORBIDDEN in the production schema; `mu2_shell` is required. Two nested scopes:

$$\texttt{A1-KERNEL-CONSISTENCY-MANIFEST}: \quad Y > 0,\ Z < 0,\ \delta_Z \le \varepsilon_Z,\ \delta_m \le \varepsilon_m,\ \delta_r \le \varepsilon_r, \tag{3}$$

$$\texttt{A1-ANALYTIC-BRANCH-MANIFEST}: \quad \texttt{A1-KERNEL-CONSISTENCY-MANIFEST} + \texttt{mu2\_shell} > 0. \tag{4}$$

Only `A1-ANALYTIC-BRANCH-MANIFEST` may be cited as the A2/A3 numerical implementation. `A1-KERNEL-CONSISTENCY-MANIFEST` also covers the condensate branch $\texttt{mu2\_shell} < 0$, but a config there must NOT be claimed to support the A2/A3 positive-gap theorems.

3. Non-vacuity and the backend certificate

The corrected δ_m discriminates: with the canonical stored fields it gives $\delta_m = 0$, and a deliberate $+0.01$ corruption of `mu2_shell` yields $\delta_m = 0.01$ (DETECTED) — assert `delta_m_is_nonvacuous_not_identically_zero`. The backend diagonal $D(k) = \texttt{r\_zero} + Z|k|^2 + Y|k|^4$ is read from the canonical reconciliation (Math426): $\texttt{r\_zero} = 0.21903$, $\texttt{mu2\_shell} = 0.00500$, $Z = -0.92528$, $Y = 1.0000$, $\texttt{q0} = 0.6801748$. This config passes BOTH manifests ($\delta_Z = 4.7 \times 10^{-11}$, $\delta_m = 2.2 \times 10^{-11}$, $\delta_r = 0$, $\texttt{mu2\_shell} > 0$). The certificate pins `config_source_sha256` and `solver_source_sha256`.

4. The legacy template fails (illustration)

The illustrative template $(r, Z, Y, q_0) = (0.35, -1, 0.50, 0.6801748)$ FAILS: $\delta_Z = 0.537$ ($q_\star^2 = 1 \neq q_0^2 = 0.463$), derived `mu2_shell` $= -0.15 < 0$ (condensate branch), and its `mu2` $= 0.35 = r$ is the forbidden alias. It is NOT citable as the A2/A3 implementation (assert `legacy_template_FAILS_and_alias_forbidden`).

5. Devil's-advocate

(α) *"Then the manifest should close — the backend passes."* VALID direction, but OPEN is retained: closure requires an operator-certified N-001 manifest with pinned $(\varepsilon_Z, \varepsilon_m, \varepsilon_r)$ and the config+solver hashes ratified. The canonical config is a certifiable CANDIDATE. (β) *"δ_m still derived?"* DISMISSED — δ_m now compares the *stored* `mu2_shell` to the identity value computed from the *stored* (`r_zero`, Z, Y); corrupting either side moves δ_m (demonstrated). (γ) *"Single tolerance fine?"* UPHELD as a defect in v1.0 — separated into $\varepsilon_Z, \varepsilon_m, \varepsilon_r$. Quantitative sanity check: $\delta_m^{\text{corrupt}} = 0.0100$ vs $\delta_m^{\text{canonical}} = 2.2 \times 10^{-11}$ (9 orders of magnitude separation).

6. Result footer

```
Result ID:               A1-PRODUCTION-KERNEL-MANIFEST (OPEN, v1.1)
Precise statement:       Stored (r_zero, mu2_shell, q0)+(Z,Y). delta_Z=|Z+2Y q0^2|,
                         delta_m=|mu2_shell-(r_zero-Z^2/4Y)|, delta_r=|r_zero-
                         (mu2_shell+Y q0^4)|, separate tol eps_Z,eps_m,eps_r.
                         CONSISTENCY: Y>0,Z<0,deltas certified. ANALYTIC-BRANCH:
                         + mu2_shell>0 (the only A2/A3-citable scope). Alias mu2=r
                         FORBIDDEN. v1.0 delta_m was identically 0 (fixed).
Scope:                   production-config certification; two nested manifests.
Dependencies:            A1-KERNEL-IDENTITY; A1-SCALAR-ANALYTIC-BRANCH.
Evidence grade:          ANALYTIC (gates) + EXECUTED (7/7, hashes pinned).
Reproduction command:    python codes/foundations/a1_kernel_checks.py
Expected output:         7/7 PASS; canonical PASSES both manifests; template FAILS.
Tier:                    T1 OPEN (canonical config is a certifiable candidate).
Falsifier:               operator-certified N-001 manifest (pinned tol + hashes)
                         closes OPEN; a config failing any delta is rejected.
No-overclaim:            schema corrected; canonical config not yet operator-
                         certified; condensate-branch configs do NOT support A2/A3.
Next required action:    Operator certifies the N-001 ANALYTIC-BRANCH manifest
                         (ratify eps_Z,eps_m,eps_r + config/solver hashes).
```